Can Day Trading Really Be Profitable? Analysis and Optimization

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1 Introduction

The debate surrounding the profitability of intraday strategies often oscillates between anecdotal success and academic skepticism. This study investigates the practical viability of day trading based on the quantitative framework established by Zarattini and Aziz. Research is developed in three progressive stages. Initially, a baseline was established by replicating the original methodology, confirming the statistical consistency of the results while accounting for high-fidelity liquidity constraints. Subsequently, the strategy was optimized by introducing advanced entry filters and the selection of high-efficiency financial instruments. Finally, the third stage involves backtesting campaign across a diverse range of assets to evaluate the strategy's robustness, **concluding with a comprehensive analysis of the empirical results**.

From a computational standpoint, this project leverages a custom-built Python simulation engine supported by a SQL-based data management architecture. This infrastructure allows for large-scale backtesting across diverse instruments, ensuring that key performance metrics are evaluated under realistic commission structures and execution models.

To ensure high-fidelity results, the market data utilized throughout the research was gathered via the IBKR API and supplemented with premium datasets from Databento. The simulation framework operates on a 5-minute intraday resolution (5-minute candles), providing the necessary granular detail to accurately model price action and execution timing while maintaining computational efficiency for large-scale data processing.

2 Theoretical Framework & Replication

2.1 Basic Strategy

Following the methodology proposed by Zarattini and Aziz, the initial phase of this study focuses on establishing a performance baseline using the Invesco QQQ Trust (QQQ) as the primary instrument. The backtesting period for this research extends from January 4, 2016, to October 31, 2025, providing a comprehensive dataset that covers nearly a decade of market evolution. To benchmark the performance generated by the proposed intraday strategy, three distinct investment approaches were compared:

- **Buy and Hold (Full-time):** A standard long-term investment benchmark representing the total return of the asset.

- **Intraday Buy and Hold:** A time-segmented approach where the position is opened at the market open (09:30) and liquidated at the market close (15:55) daily. This benchmark serves to isolate the returns generated strictly during regular trading hours.

- **Basic Strategy (Naïve Momentum):** A simplified version of the core strategy which enters a long position at the first available price if the opening 5-minute candle is positive, or a short position if it is negative. All positions are closed at 15:55.

For this preliminary replication, the backtest assumes the execution of a single share per trade, ignoring initial capital constraints and transaction costs. This "raw" performance analysis is intended to validate the underlying momentum logic before introducing complex filters and slippage models.

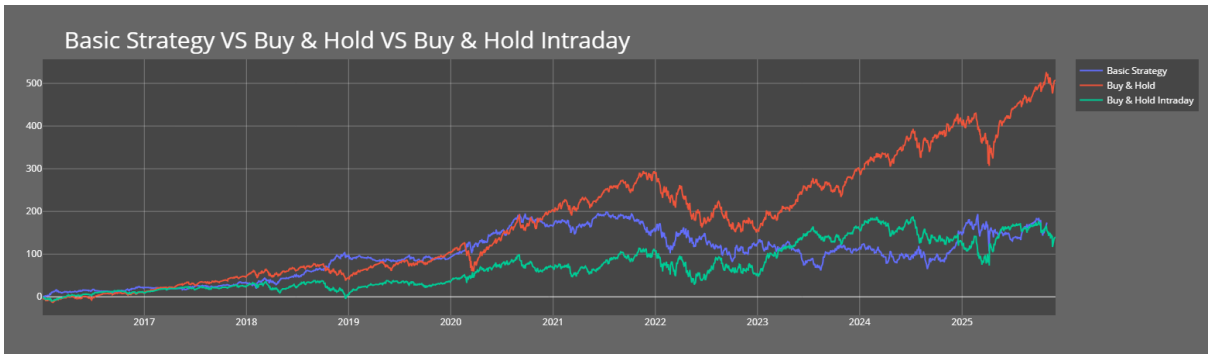


Figure 1: Performance Comparison

Observations: The strategy's signal appears to have undergone a significant deterioration starting in 2021; nevertheless, its performance remains broadly in line with an intraday Buy & Hold Intraday approach.

2.2 Test 1

The second step in the replication involves the integration of structured exit logics, as defined in the original paper. To transition from a simple time-based exit to a risk-managed framework, the following parameters were implemented:

- **Stop Loss:** For long positions, the Stop Loss is set at the *low* of the 09:30 opening candle. Conversely, for short positions, the Stop Loss is placed at the *high* of the same candle.

- **Fixed Take Profit:** The profit-taking target is set at a 10:1 Risk/Reward ratio (10 RR). Defining the risk unit (R) as the distance between the entry price and the Stop Loss.

The results are as follows:

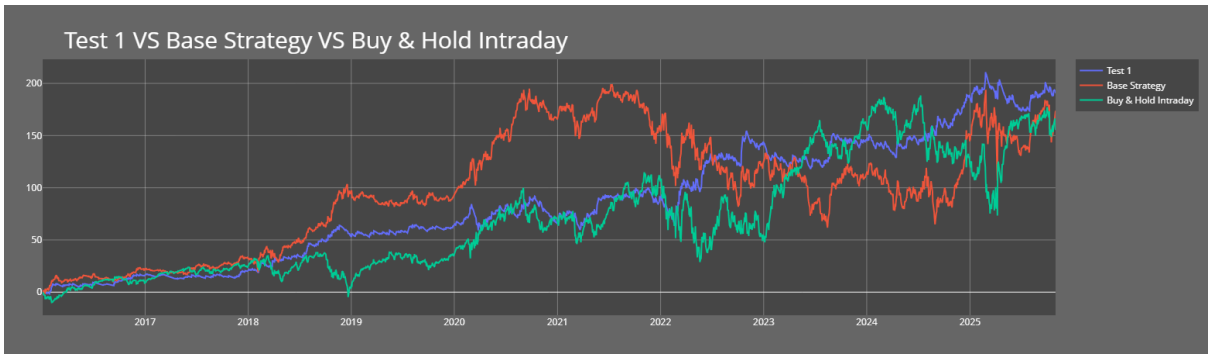


Figure 2: Performance Comparison

Observations: The results have improved, placing the strategy ahead of the intraday Buy & Hold benchmark. Moreover, the equity curve exhibits a much smoother and more consistent trajectory.

2.3 Test 1 Money Management

In this section, specific money management rules and transaction costs are implemented to transition toward a realistic representation of the strategy's capitalization.
We assume an initial capital of $25,000, with a commission structure of $0.0035 (per part) per share, in accordance with the Interactive Brokers (IBKR) tiered pricing model. The number of shares to be traded (Position Size) is dynamically determined by the following formula:

$$\text{Size} = \min\left(\frac{\text{AccountValue} \times 0.01}{\text{Risk}}, \frac{\text{AccountValue} \times 4}{\text{Opening Price}}\right) \tag{1}$$

This approach ensures that the risk per trade is capped at 1% of the account equity, while simultaneously respecting a maximum intraday leverage of 4:1, as per standard U.S. margin requirements.

Figure 3: Performance Comparison

Observations: The strategy appears to achieve a significantly higher return compared to a simple Buy & Hold approach. However, this outperformance is largely driven by

the fact that the strategy trade an increasingly larger number of units. It is therefore essential to verify the actual size of the traded positions to ensure the strategy remains practically feasible and is not undermined by slippage constraints.

The evolution of the position size for each individual trade, calculated according to the previously defined money management rules, is illustrated in the following chart:

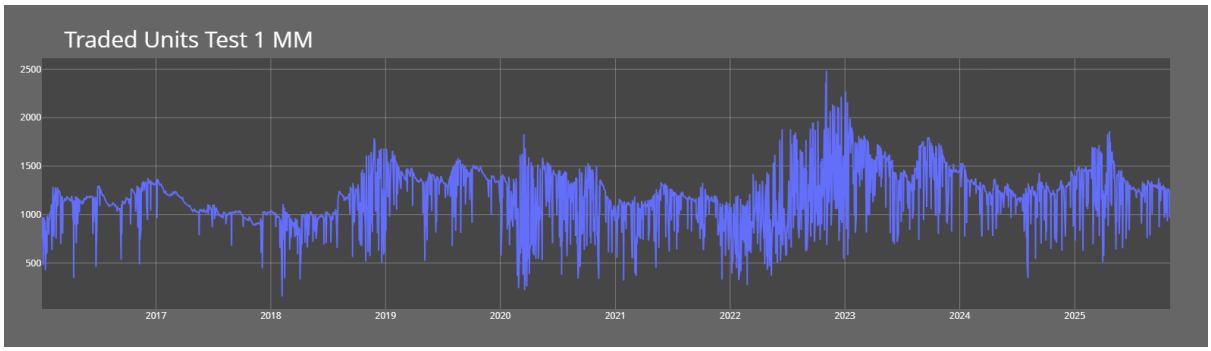


Figure 4: Number of Shares Traded per Operation

The application of the dynamic position sizing model resulted in an average exposure of 1172.89 shares per operation.

Observations: As expected, the strategy's position size grows to very high levels. Observing QQQ during regular trading hours, the Top of Book - that is, the best bid and ask levels - typically displays only 1,000 to 2,000 shares. Consequently, executing larger orders at market results in slippage. The next step is to simulate this slippage in our model.

2.4 Slippage Simulation Model

To assess execution quality under adverse market conditions, we construct a conservative slippage simulation model based on a synthetic order book. The model is designed to approximate *worst-case execution scenarios* by combining empirical market observations with a hierarchical random variable framework. The resulting weighted average fill price accounts for the gradual deterioration of liquidity across successive price levels.

Order Execution Logic

Crucially, the slippage model is selectively applied based on the order type:

- **Market and Stop Orders:** Slippage is strictly enforced for entry signals (Market orders) and Stop Loss triggers (Stop-Market orders). These orders prioritize execution speed over price, effectively "crossing the spread" and consuming the available liquidity across multiple book levels.

- **Limit Orders:** Conversely, Take Profit targets are modeled as Limit orders. These are assumed to be resting in the book, providing liquidity rather than consuming it; therefore, they are executed at the exact target price without incurring slippage.

Order Book Structure

The simulated order book is organized into discrete price levels, each characterized by a random variable governing the available liquidity. Execution proceeds sequentially from the best available price (Level 0) through deeper levels until the entire order size Q is filled.

Level 0 — Top of Book: The liquidity at the best price level is modeled as a truncated normal distribution, ensuring a strictly positive quantity:

$$q_0 \sim \mathcal{TN}\big(\mu_0,\, \sigma_0^2;\, q_{\min},\, +\infty\big)$$

where μ_0 and σ_0 are calibrated from market observations, and $q_{\min}$ represents the minimum observable quantity at the top of the book. The realized draw q_0 anchors the liquidity profile of all subsequent levels.

Level 1: The available quantity at Level 1 is expressed as a random fraction of q_0:

$$q_1 = \lfloor U_1 \cdot q_0 \rceil, \qquad U_1 \sim \mathcal{U}(0.5,\ 0.8)$$

reflecting a reduction to between 50% and 80% of the top-of-book liquidity.

Level 2: Liquidity deteriorates further at Level 2:

$$q_2 = \lfloor U_2 \cdot q_0 \rceil, \qquad U_2 \sim \mathcal{U}(0.3,\ 0.5)$$

corresponding to 30% to 50% of q_0.

Levels $\ell \geq 3$: For all deeper levels, liquidity is modeled as a uniformly distributed fraction in the lower tail:

$$q_\ell = \lfloor U_\ell \cdot q_0 \rceil, \qquad U_\ell \sim \mathcal{U}(0.1,\ 0.3), \qquad \ell \geq 3$$

Execution Price by Level

At each level ℓ, the execution price deviates from the theoretical price p^* by a fixed tick increment δ per level, modeling the cost of walking the book:

$$p_\ell = \begin{cases} p^* + \ell\,\delta & \text{(Buy order)} \\ p^* - \ell\,\delta & \text{(Sell order)} \end{cases}$$

where δ denotes the tick size of the instrument.

Weighted Average Fill Price

$\tilde{q}_\ell$ denote the quantity filled at level ℓ and ℓ^* indicates the level at which the order is fully executed. The volume-weighted average fill price is:

$$\bar{p} = \frac{\sum_{\ell=0}^{\ell^*} \tilde{q}_\ell \cdot p_\ell}{Q}$$

The realized slippage per share relative to the theoretical price is then:

$$\varepsilon = \bar{p} - p^* \quad \text{(Buy)}, \qquad \varepsilon = p^* - \bar{p} \quad \text{(Sell)}$$

2.5 Test 1 Slippage Simulation

The performance of the strategy, accounting for dynamic slippage, is as follows:

$$q_0 \sim \mathcal{TN}(750,\, 300;\, 100,\, +\infty)$$

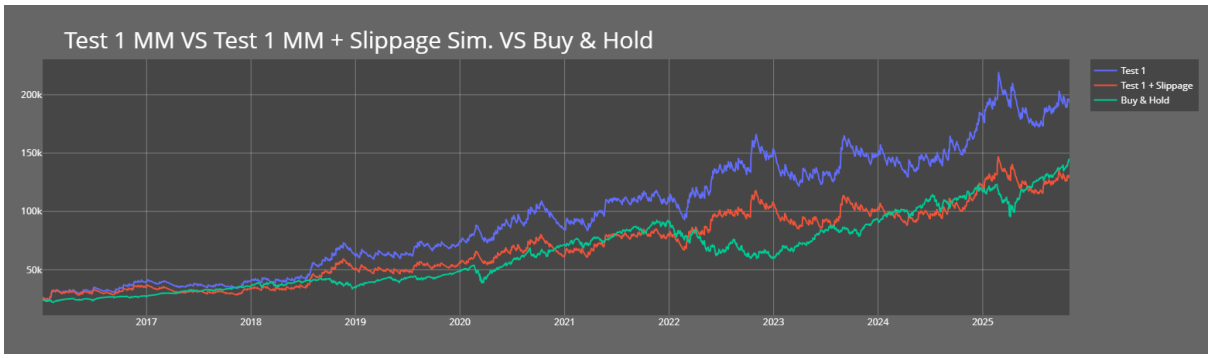


Figure 5: Performance Comparison

Observations: As illustrated in the figure, the introduction of a dynamic slippage model reveals that the strategy's performance does not significantly deviate from the benchmark. This suggests that, if deployed in a live trading environment, the strategy would lack a genuine competitive edge. Consequently, subsequent sections will explore the transition to high-efficiency instruments, such as MNQ, which offer superior liquidity to mitigate slippage and transaction costs. Furthermore, as futures contracts, instruments like MNQ provide extended trading hours, which will be leveraged to enhance signal quality and overall system robustness.

3 Improvements

3.1 MNQ Test 1

The primary objective of this section is to evaluate the scalability and performance of the strategy when transitioned from the QQQ ETF to **Micro E-mini Nasdaq-100 (MNQ) futures**. For this implementation, execution costs are modeled according to the Interactive Brokers (IBKR) Tiered commission structure, set at \$0.25 per side (\$0.50 round-turn per contract). To maintain consistency with the previous liquidity analysis, the Top-of-Book volume (q_0) is simulated using a truncated normal distribution:

$$q_0 \sim \mathcal{TN}(6, 2; 2, +\infty) \tag{2}$$

Futures Data: Raw futures data were obtained via DataBento. The dataset was reconstructed to form a continuous series based on the most liquid *front-month* contracts. Notably, price series are not back-adjusted; given the intraday nature of the strategy, the absence of overnight holdings eliminates the need for roll-gap adjustments. It is important to note that the available historical data for MNQ futures commences on May 6, 2019.
Figure 6 illustrates the comparative performance trajectory between the MNQ and QQQ implementations.

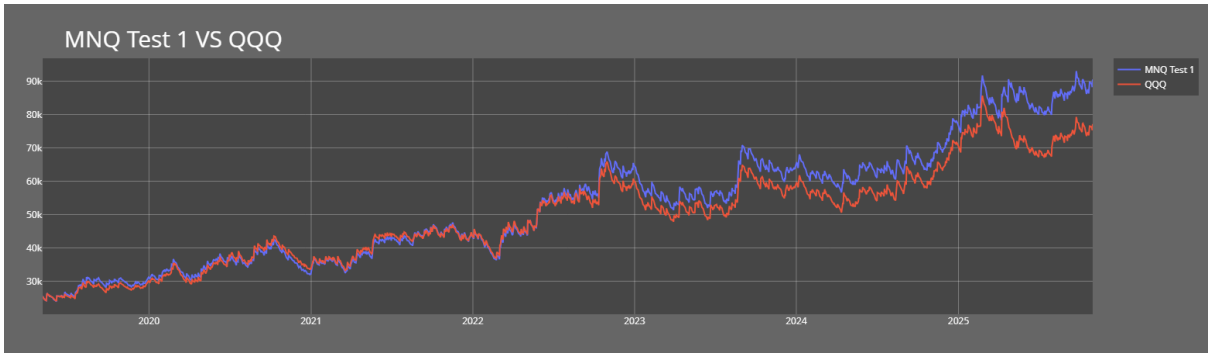


Figure 6: Performance Comparison

Observations: The MNQ-based strategy exhibits a compounding performance advantage over the QQQ benchmark. While minor stochastic variations in trade execution exist, the primary driver of this outperformance is the significant reduction in *cost drag*. A granular analysis of the **Average Profit per Unit** reveals a stark contrast:

- **MNQ:** 7.02 \$/unit
- **QQQ:** 0.077 \$/unit

This indicates a substantially wider operating margin for the futures-based approach, as the commission-to-notional ratio is more favorable for MNQ contracts than for the equivalent ETF share volume.

Furthermore, the average position size for the MNQ test was **6.13 contracts**. Given that this mean aligns closely with the modeled liquidity ($\mu = 6$), the strategy consistently operates within the first level of the order book, resulting in near-optimal execution.

While the QQQ implementation experienced slightly higher slippage due to larger share blocks frequently exhausting the Top-of-Book volume, but this execution lag was not the primary detractor of performance.

The distribution of traded units for the MNQ test 1 is presented in Figure 7.

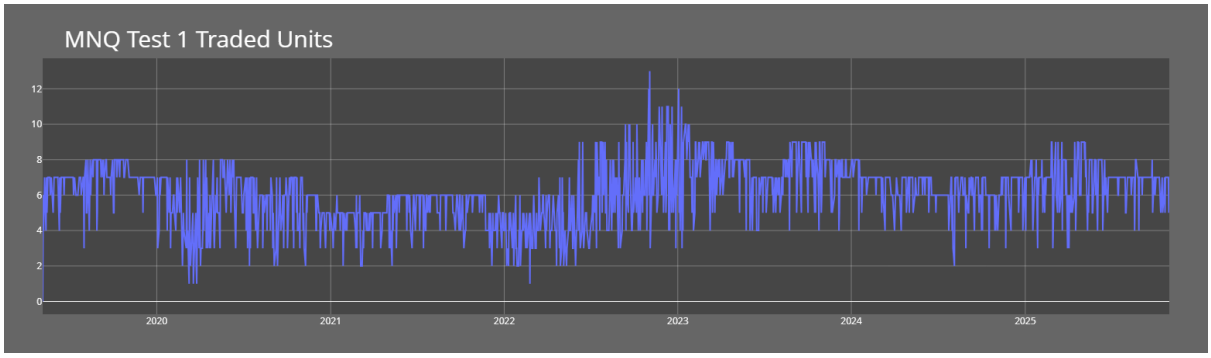


Figure 7: Number of Contracts Traded per Operation

3.2 MNQ Test 2

The subsequent phase of the research focuses on refining the entry signal and optimizing the stop-loss placement to improve the strategy's risk-adjusted returns. In **MNQ Test 2**, the lookback window is extended to include the 09:25 candle (five minutes prior to the official market open). The signal is determined by the cumulative price action from the 09:25 open to the 09:30 close: a positive net return triggers a *long* position, while a negative return triggers a *short* position.

Refined Stop-Loss Logic: The stop-loss mechanism maintains its original logic but is now calculated across the combined range of both candles. Specifically, for long positions, the stop is placed at the minimum low of the 09:25 - 09:30 period; for short positions, it is set at the maximum high of the same interval.

The comparative performance between Test 1 and Test 2 is illustrated in Figure 8.

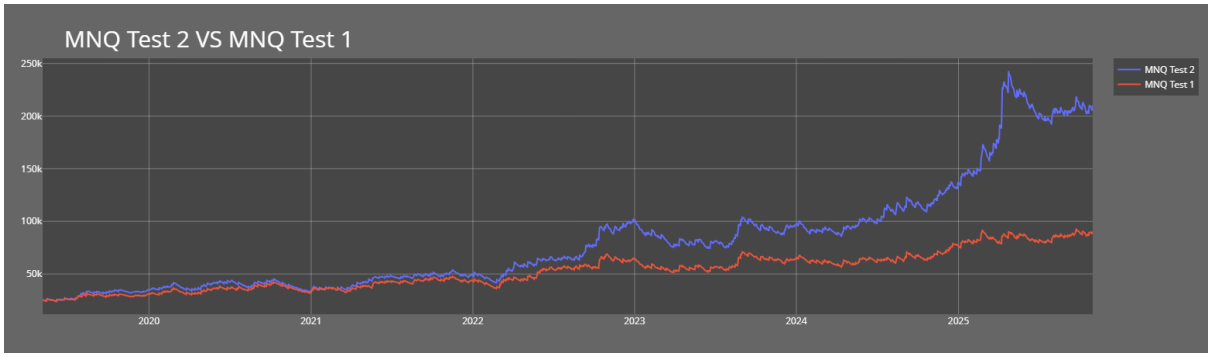


Figure 8: Performance Comparison

Observations: The implementation of the 5-minute lookback window yielded an extraordinary **132.5% increase** in net performance compared to MNQ Test 1. The final metrics highlight a significant enhancement in the strategy's expectancy:

- **MNQ Test 1:** Final Equity = \$90,558; Average Profit per Unit = \$7.01
- **MNQ Test 2:** Final Equity = \$209,996; Average Profit per Unit = \$13.11

3.3 MNQ Test 3

In this phase, a dynamic risk management system is integrated to optimize trade handling based on unrealized PnL and R-multiple thresholds. The system monitors open positions at the close of each candle and adjusts the stop-loss order according to the following rules:

- **Break-Even Trigger:** Once the unrealized profit reaches a $1R$ threshold, the stop-loss is moved to the average entry price, effectively rendering the trade risk-free.
- **Trailing Logic:** If the unrealized profit hits a $6R$ level, the stop-loss is advanced to lock in a minimum profit of $5R$. This mechanism is designed to prevent significant drawdowns on highly profitable positions due to sudden market reversals.

The results of the active risk management implementation are presented in Figure 9.

Figure 9: Performance Comparison

Observations: The introduction of a dynamic stop-loss management does not yield a significant improvement in overall performance; however, a notable reduction in strategy volatility is observed. A detailed examination of this reduced variability will be conducted in the following section utilizing a **linear regression analysis**.

3.4 Linear Regression Analysis

To objectively evaluate the quality of the equity growth between MNQ Test 2 and MNQ Test 3, a comparative regression analysis was performed. To ensure statistical validity, two essential preprocessing steps were implemented:

- **Log-Transformation:** The equity series were converted to log-space (log(Equity)) to mitigate heteroskedasticity, ensuring that the variance of the residuals remains constant over time.

- **Time Indexing:** The temporal variable was indexed as the "number of trades," providing a consistent independent variable for the OLS model.

Using the `statsmodels` library, a linear regression was fitted for both tests. The visual results of the trend-line fitting are shown in Figure 10.

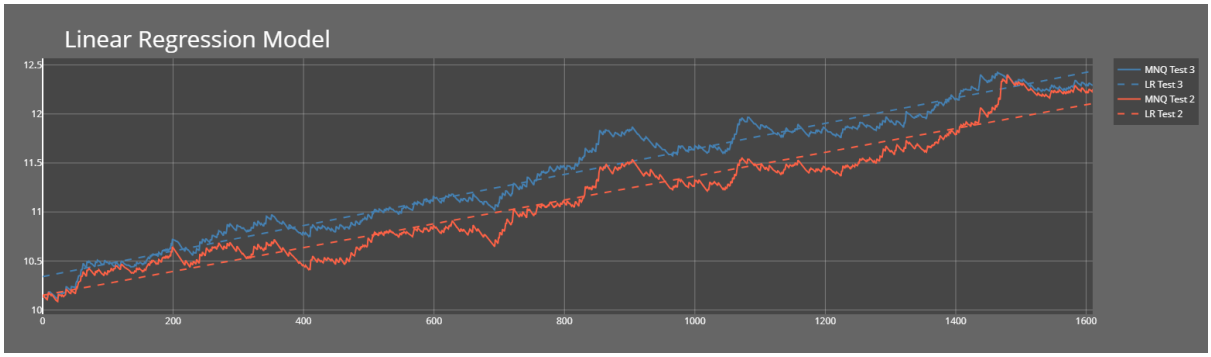


Figure 10: OLS Regression Analysis

The statistical metrics derived from the models provide a clear indication of the structural improvements introduced in Test 3:

Strategy Version	**R-Squared (R^2)**	**RMSE**
MNQ Test 2 (Standard)	0.94	0.14
MNQ Test 3 (Active Risk)	**0.9666**	**0.11**

Observations: The results demonstrate that MNQ Test 3 possesses a higher explanatory power, with an R^2 of **0.9666**. This indicates that 96.66% of the equity variation is explained by the linear progression of trades, compared to 94% for Test 2.

Furthermore, the **Root Mean Square Error (RMSE)** - which represents the average deviation of the log-equity from its ideal linear trend - decreased from 0.14 to 0.11. This 21% reduction in residual volatility confirms that the active risk management system significantly dampens drawdowns and promotes a more "straight-line" capital appreciation. In conclusion, MNQ Test 3 is not only more profitable but exhibits higher statistical reliability and lower path-dependency risk.

4 Final Performance Analysis

This concluding section presents the empirical results of the strategy across the tested assets. The performance is evaluated using a comprehensive suite of risk and return metrics.

4.1 Performance Metrics

To provide a granular assessment of the strategy's expectancy and risk profile, the following metrics are reported:

- **Profit & Loss per Contract:** The net monetary gain or loss attributed to a single traded contract.
- **Win Rate:** The percentage of trades resulting in a positive net return.
- **Expected Value per contract (E_u):** The average profit generated for each contract traded, accounting for the probabilistic distribution of outcomes.
- **Expected Return per Trade (%):** This metric represents the strategy's mathematical "edge" expressed as a percentage of the notional exposure per execution. It quantifies the average yield generated by a single trade.
- **Compound Annual Growth Rate (CAGR):** The geometric progression ratio that provides a constant rate of return over the investment period, calculated as:

$$\text{CAGR} = \left(\frac{V_{final}}{V_{initial}}\right)^{\frac{1}{t_{\text{years}}}} - 1 \tag{3}$$

- **Total Number of Trades:** The sample size of the execution history.
- **Average Traded Units (20-period MA):** A 20-period moving average of the position size.
- **Maximum Drawdown (MaxDD):** The largest peak-to-trough decline observed in the equity curve, representing the worst-case historical loss.
- **Average Drawdown (Average DD):** The mean depth of all retracements from historical equity peaks.
- **Average Drawdown Duration (DD Duration):** The mean time to recovery, measured by the number of trades required to reach a new equity high.

4.2 MNQ

Instrument: Micro E-mini Nasdaq-100 Futures (MNQ)
Asset Class: Equity Index Futures
Backtest Period: 2019-05-06 to 2025-11-26
Initial Equity: $25.000
Slippage Modeling:

$$q_0 \sim \mathcal{TN}(6, 2; 2, +\infty) \tag{4}$$

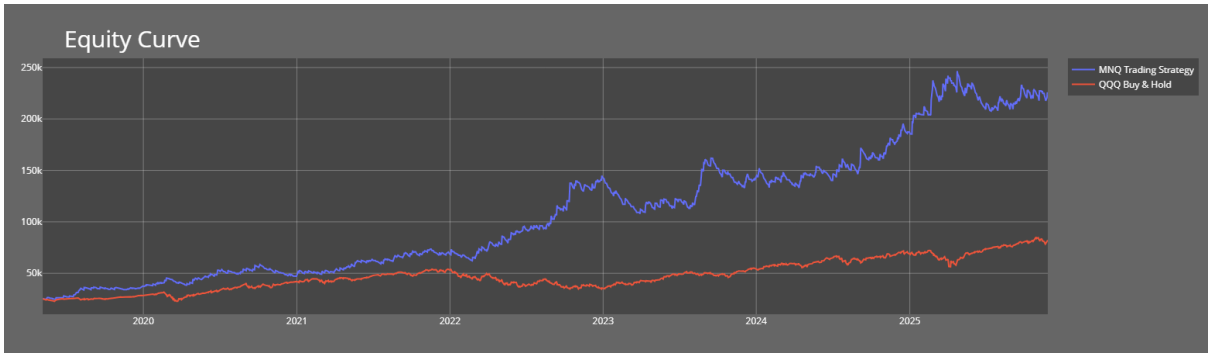


Figure 11: MNQ Equity Curve

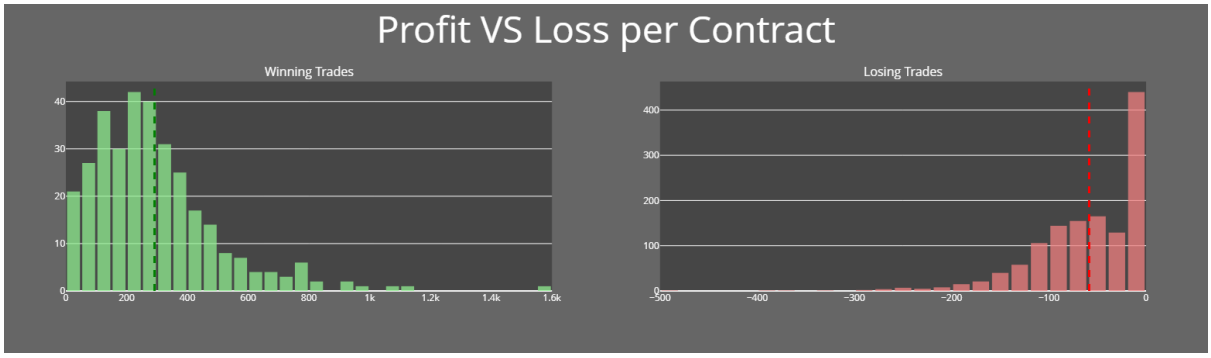


Figure 12: MNQ Profit & Loss per Contract

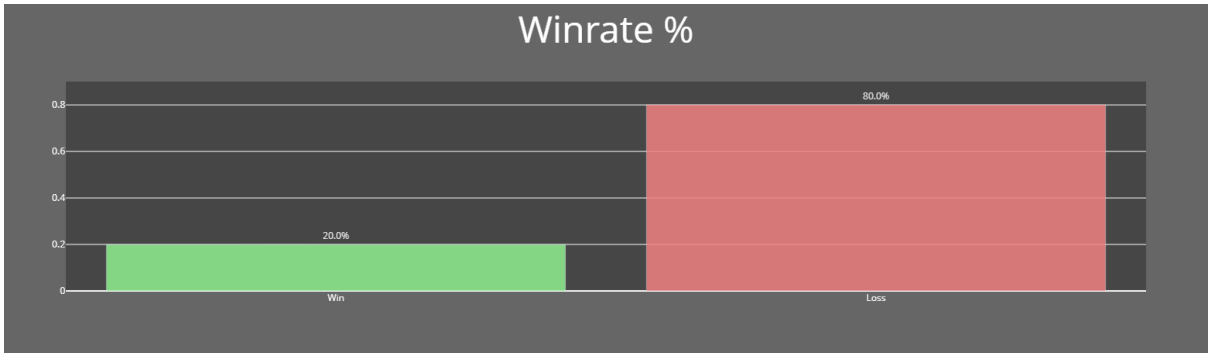


Figure 13: MNQ Winrate

MNQ Trading Strategy - $ EV per Unit

11.38$

MNQ Trading Strategy - % EV per Trade

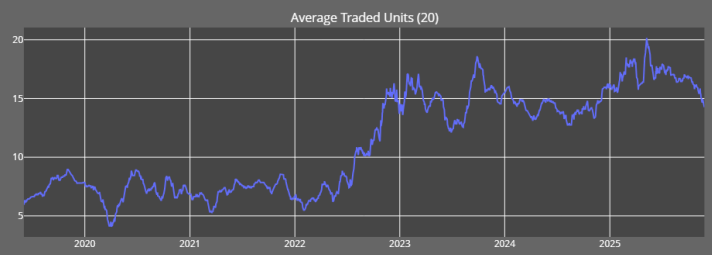
0.15%

Figure 14: MNQ EV

MNQ Trading Strategy - % CAGR

38.94%

Figure 15: MNQ CAGR



Number of Trades

1628

Figure 16: MNQ Average Traded Contarcts & Number of Trades

Drawdown Statistics

MNQ Trading Strategy - Max DD	Average DD	DD Duration (Trades)
−28.71%	−3.85%	15.1
QQQ Buy & Hold - Max DD	**Average DD**	**DD Duration (Trades)**
−35.68%	−2.97%	15.2

Figure 17: MNQ Drawdown Statistics

4.3 NQ

Instrument: E-mini Nasdaq-100 Futures (NQ)
Asset Class: Equity Index Futures
Backtest Period: 2015-01-02 to 2025-11-26
Initial Equity: $100,000
Slippage Modeling:

$$q_0 \sim \mathcal{TN}(8, 3; 3, +\infty) \tag{5}$$

Figure 18: NQ Equity Curve

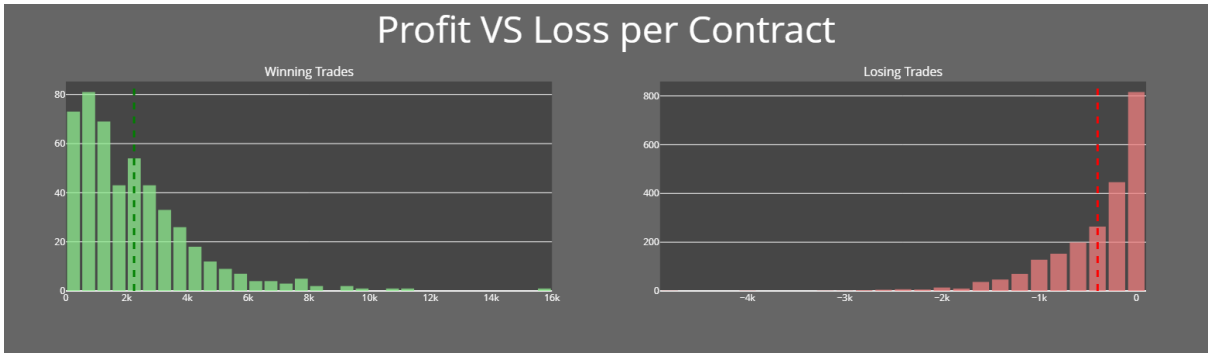


Figure 19: NQ Profit & Loss per Contract

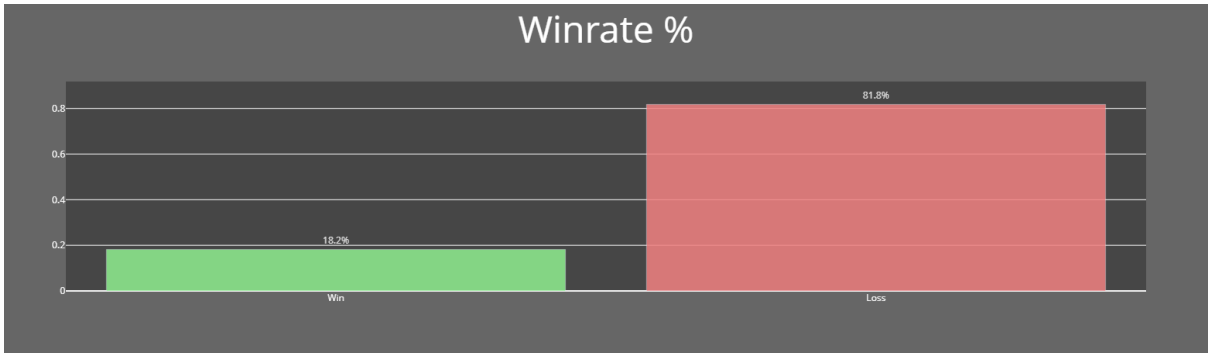


Figure 20: NQ Winrate

NQ Trading Strategy - $ EV per Unit

82.85$

NQ Trading Strategy - % EV per Trade

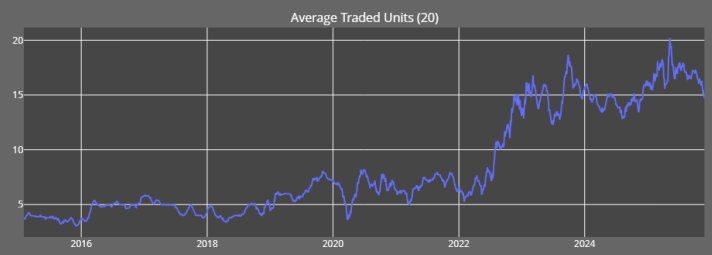
0.13%

Figure 21: NQ Expected Value

NQ Trading Strategy - % CAGR

32.77%

Figure 22: NQ CAGR



Number of Trades

2701

Figure 23: NQ Average Traded Contracts & Number of Trades

Drawdown Statistics

NQ Trading Strategy - Max DD	Average DD	DD Duration (Trades)
−27.09%	−4.00%	19.3
QQQ Buy & Hold - Max DD	Average DD	DD Duration (Trades)
−35.64%	−2.72%	14.7

Figure 24: NQ Drawdown Statistics

4.4 ES

Instrument: E-mini S&P 500 Futures (ES)
Asset Class: Equity Index Futures
Backtest Period: 2015-01-02 to 2025-11-26
Initial Equity: $100,000
Slippage Modeling:

$$q_0 \sim \mathcal{TN}(9, 3; 3, +\infty) \tag{6}$$

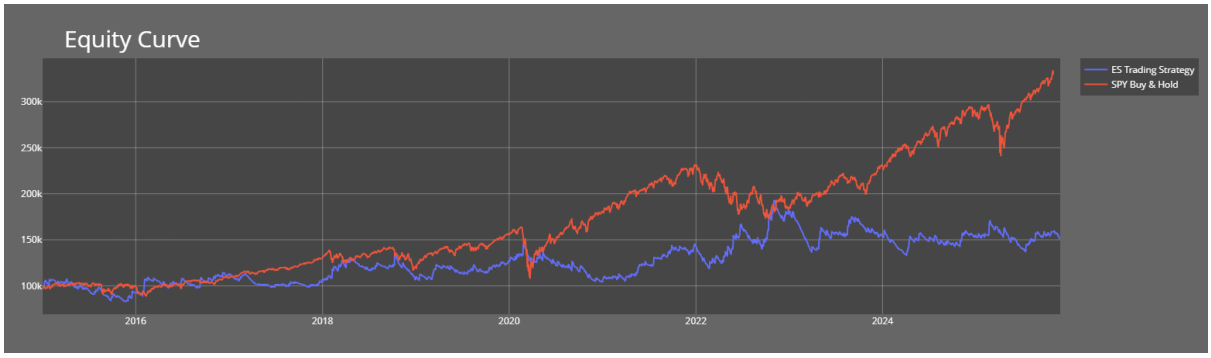


Figure 25: ES Equity Curve

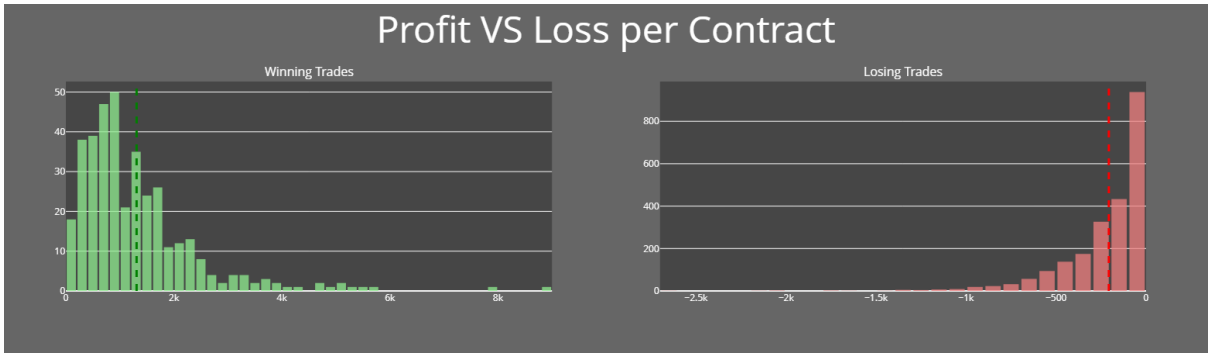


Figure 26: ES Profit & Loss per Contract

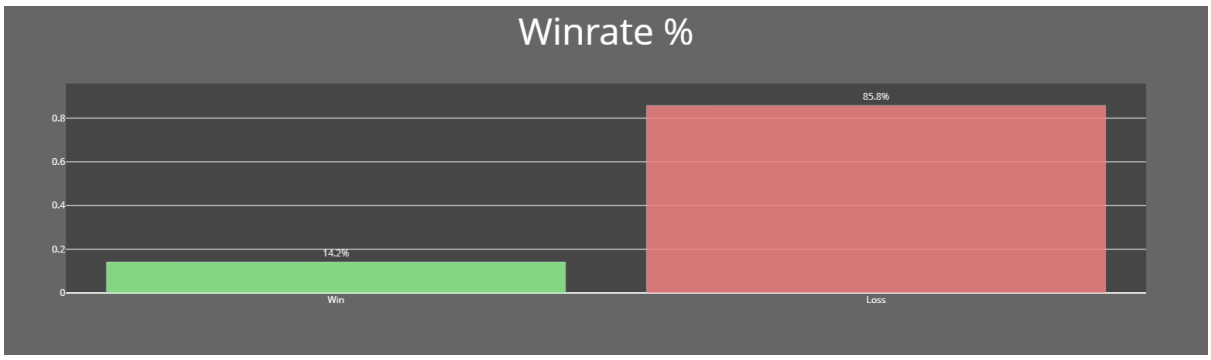


Figure 27: ES Winrate

ES Trading Strategy - $ EV per Unit

8.09$

ES Trading Strategy - % EV per Trade

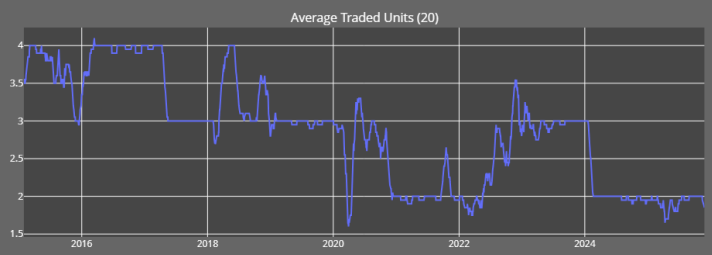
0.02%

Figure 28: ES Expected Value

ES Trading Strategy - % CAGR

3.81%

Figure 29: ES CAGR



Number of Trades

2643

Figure 30: ES Average Traded Contracts & Number of Trades

Drawdown Statistics

ES Trading Strategy - Max DD	Average DD	DD Duration (Trades)
−30.93%	−7.60%	84.2
SPY Buy & Hold - Max DD	Average DD	DD Duration (Trades)
−34.25%	−1.80%	13.9

Figure 31: ES Drawdown Statistics

4.5 YM

Instrument: E-mini Dow Jones Industrial Average Futures (YM)
Asset Class: Equity Index Futures
Backtest Period: 2015-01-02 to 2025-11-26
Initial Equity: $100,000
Slippage Modeling:

$$q_0 \sim \mathcal{TN}(6, 2; 2, +\infty) \tag{7}$$

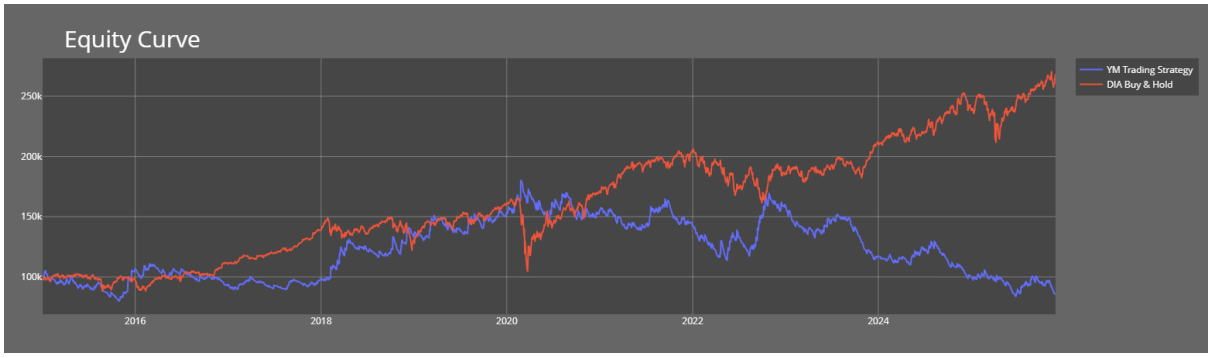


Figure 32: YM Equity Curve

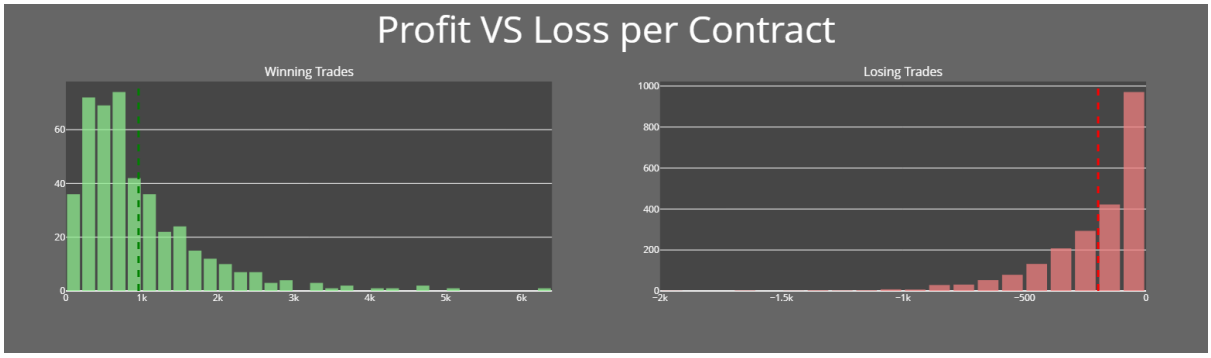


Figure 33: YM Profit & Loss per Contract

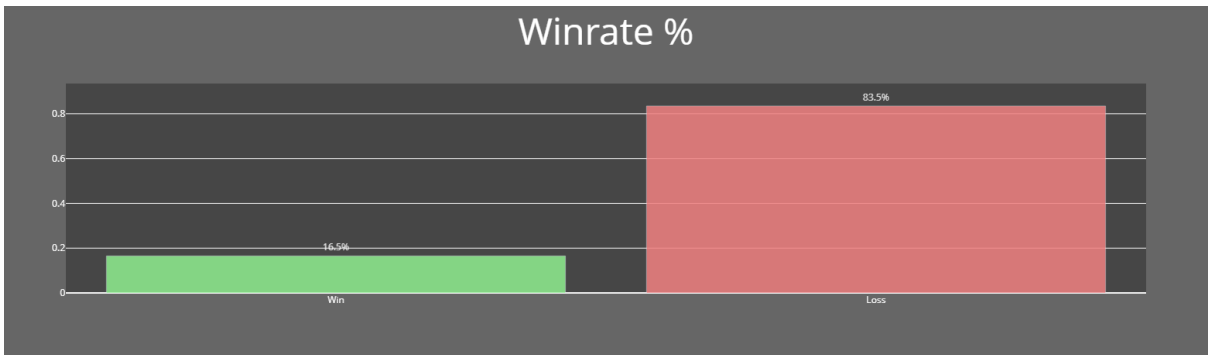


Figure 34: YM Winrate

YM Trading Strategy - $ EV per Unit

−6.35$

YM Trading Strategy - % EV per Trade

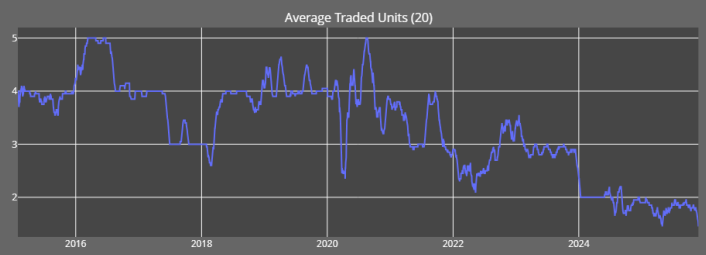
0.00%

Figure 35: YM Expected Value

YM Trading Strategy - % CAGR

−1.40%

Figure 36: YM CAGR



Number of Trades

2690

Figure 37: YM Average Traded Contracts & Number of Trades

Drawdown Statistics

YM Trading Strategy - Max DD	Average DD	DD Duration (Trades)
−53.77%	−7.96%	110.8
DIA Buy & Hold - Max DD	**Average DD**	**DD Duration (Trades)**
−37.14%	−2.06%	18.5

Figure 38: YM Drawdown Statistics